6/8/24

EXPERIMENT-4

STUDY OF LOCAL AREA NETWORK (LAN) USING A SWITCH AND ETHERNET CABLES

AIM:

To setup & configure LAN (Local Area Network) using a switch & Ethernet cables in an lab.

WHAT IS A LAN?

A Local Area Network (LAN) refers to a network that connects devices within an limited area, such as office building, school or home. It enables users to share resources, including data, printers & internet access. A LAN switch serves as the primary connecting device, managing & directing communications within the local network.

PROCEDURE:

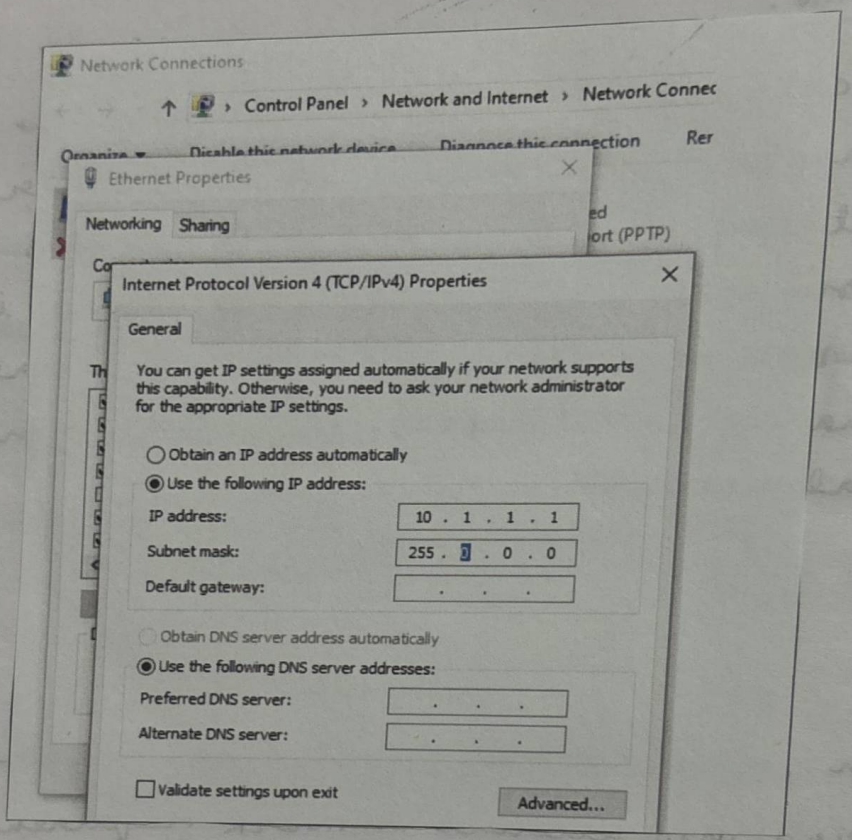
STEP 1: Plan & Design an appropriate network topology taking into account network requirements & equipment location.

STEP 2: You can take 4 computers, a switch with 8, 16, 24 ports which is sufficient for network of these sizes & 4 ethernet cables.

STEP 3: Connect your computer to network switch via an ethernet cable, which is as simple as plugging one end of the ethernet cable into your computer & the other end into your network switch.

STEP 4: Assign IP address to your PC's:

1) Log on to the client computer as administrator/ owner.
2) Click Network & Internet Connections.
3) Right Click LA Connections / Ethernet → Go to Properties → select Internet Protocol → Click on Properties → Select use the following ip address option & assign ipaddress.


Network Connections
Control Panel › Network and Internet › Network Connec
Ethernet Properties
Networking
Sharing
Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address:
10 . 1 . 1 . 1
Subnet mask:
255 . 0 . 0 . 0
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...

→ PC1 : 10.1.1.1 , subnet mask 255.0.0.0

→ PC2 : 10.1.1.2, subnet mask 255.0.0.0

→ PC3 : 10.1.1.3, subnet mask 255.0.0.0

→ PC4 : 10.1.1.4 , subnet mask 255.0.0.0

STEP 5: Configure the switch by connecting it to a computer, accessing it, web interface & assigning the switch IP as 10.1.1.5 with 255.0.0.0

RESULT:

Thus, a local area network (LAN) was successfully setup & configured using a network switch & ethernet cables.